Choosing a pillow for neck pain: what the trials actually show

An evidence summary built only from randomized trials, systematic reviews and ergonomic studies in peer-reviewed journals. No brand rankings, no sponsored reviews. Prepared 10 Nov 2026 for personal and pharmacist reference.

The bottom line, before the details

No sleep-medicine body names a brand. What exists is a small trial literature (mostly physiotherapy and ergonomics journals) that compares pillow *materials, heights and shapes*. That is what this document summarizes.

Material: solid **latex (natural rubber)** has the most consistent evidence for fewer waking neck symptoms and better comfort (Gordon 2009, Gordon 2010, meta-analysis Pang 2021, review Radwan 2021). A pocket-**spring** pillow beat education alone in one good RCT (Vanti 2019).

Avoid feather/down: the worst performer in every head-to-head field trial (roughly 2 to 2.6 times the odds of waking stiffness, headache and scapular/arm pain versus the sleeper's own pillow; one-third of participants quit it).

Shape is over-sold: the foam contour ("cervical" wave) pillow did no better than a plain foam pillow. Polyester and plain foam performed about the same as people's existing pillows.

Height: for side sleepers, a loaded height of about **10 cm** was the most comfortable and lowered neck-muscle activity versus a low (5 cm) pillow; for back sleepers the best data point to roughly 7 to 10 cm. Reviews give 7 to 11 cm (unloaded, central section) as the supported range.

A pillow is an adjunct, not a treatment. In the only factorial RCT, a neck-support pillow alone was no better than placebo; combined with physiotherapist-led neck exercise it gave the best result (Helewa 2007).

1. What the evidence says, feature by feature

1.1 Material

The best field data come from Gordon and colleagues in South Australia: 106 healthy side-sleepers tried their own pillow plus five trial pillows (polyester, regular foam, foam contour, feather, latex), one week each with washout weeks between, recording symptoms every morning (Manual Therapy 2009; Journal of Pain Research 2010). Latex had the highest completion rate (97%), lower odds of waking headache than polyester (OR 0.3, 95% CI 0.1 to 0.8) and performed better than the sleepers' own pillows on waking pain, sleep quality and comfort in most comparisons. Feather had the most waking symptoms and the most dropouts (67% completion): versus own pillow, OR 2.0 for any stiffness, 2.4 for headache and 2.6 for scapular/arm pain. The authors' wording: "latex pillows can be recommended over any other type" and feather pillows "should not be recommended."

The 2021 meta-analysis by Pang, Tsang and Fu (35 articles screened; 9 higher-quality studies, 555 participants, mean PEDro 6.6/10) pooled rubber (latex) and spring pillows against feather and usual pillows in people with chronic neck pain: neck pain SMD -0.26 ($p < 0.001$), waking symptoms SMD -0.23 ($p < 0.001$), neck disability SMD -0.51 ($p = 0.02$), satisfaction SMD +1.14 ($p < 0.001$). Sleep quality did *not* change (SMD 0.05, $p = 0.70$). The Radwan 2021 review (11 controlled studies, 309 participants) reached the same conclusion: latex has "the strongest evidence" for comfort, less neck pain and better function.

Caveat for degenerative necks. A later pilot RCT in 92 side-sleepers with radiologically confirmed cervical spondylosis and daily waking symptoms (Gordon 2019) compared latex and polyester for 28 days each: neither pillow changed any outcome, and latex was tolerated worse. The authors estimate 400+ symptomatic participants would be needed to show an effect. So the latex advantage is established in the

general side-sleeping population with waking symptoms, and unproven in established spondylosis.

1.2 Shape and "cervical" contours

In the Gordon trials the contoured foam pillow was no better than regular foam on any outcome, and both were associated with longer stiffness episodes than the sleeper's own pillow (Gordon 2010). The older Swedish comparison of six neck-support pillows (Persson & Moritz 1998, 55 participants, 3 weeks) found people preferred a pillow that was soft, not too high, washable, and had *firm support under the cervical curve*: the design with two firmer cores supporting the neck was rated best. Read together: support under the neck matters; a moulded wave shape does not add to it.

1.3 Height (loft)

Sacco 2015 measured surface EMG of sternocleidomastoid and trapezius in 21 adults lying on their side on 5, 10 and 14 cm foam pillows: the 5 cm pillow produced the highest down-side middle-trapezius activity and the lowest comfort; 10 cm was rated most comfortable ($p < 0.001$) and did not differ from 14 cm on muscle activity. In the supine position, radiographs favoured 10 cm over 0 or 20 cm for neutral cervical alignment, and comfort ratings favoured 7 cm over 3, 11 or 15 cm (studies summarized in Lei 2021). That review's honest conclusion is that "there remain no firm conclusions about the optimal pillow height"; anthropometrics such as BMI and shoulder width predicted preferred height poorly. Practical reading: side sleepers need roughly the ear-to-shoulder distance filled once the pillow compresses, which for most adults lands near 10 cm loaded; back sleepers need less.

1.4 Spring and water pillows

Vanti 2019 (Physical Therapy; PEDro 8/10) randomized 70 adults with chronic non-specific neck pain to a viscoelastic foam pillow with 60 independent springs or to ergonomic education for 4 weeks, then crossed over. The spring pillow reduced neck pain by 8.7 points on a 0 to 100 scale (95% CI 2.6 to 14.7), thoracic pain by 8.4 and headache by 16.0 versus education; shoulder pain and sleep quality did not differ. Lavin 1997 found a water-filled pillow beat a roll pillow and usual pillow for morning pain and sleep quality in 41 patients, but the trial is low quality (PEDro 3/10, unblinded, no concealed allocation).

1.5 Pillow versus exercise

Helewa 2007 randomized 151 adults with chronic neck pain in a 2 x 2 factorial design: neck-support pillow, physiotherapist-led neck exercise, both, or placebo (hot/cold packs and massage). At 12 weeks the Northwick Park score was 18.6 (placebo), 21.5 (pillow alone), 20.1 (exercise alone) and 14.1 (both). Neither main effect was significant; the interaction was ($p = 0.029$). The pillow helped only when paired with exercise.

2. Evidence table

Study	Design and sample	Compared	Key result
Gordon 2009, 2010	Blinded randomized field trial; 106 healthy side-sleepers; 6 pillows x 1 week	Own, polyester, foam, foam contour, feather, latex	Latex best (fewer waking symptoms, 97% completion); feather worst (OR 2.0 to 2.6 for symptoms, 67% completion); contour = plain foam
Gordon 2019	Pilot double-blind RCT; 92 side-sleepers with cervical spondylosis; 28 days each	Latex vs polyester	No significant change in any outcome; latex tolerated worse; needs 400+ participants
Pang 2021	Systematic review and meta-analysis; 9 studies, 555 participants	Rubber/spring vs feather/usual	Neck pain SMD -0.26; disability SMD -0.51; satisfaction SMD +1.14; sleep quality no effect
Radwan 2021	Systematic review; 11 controlled studies, 309 participants	Material, height, shape, surface	Moderate evidence: latex material; 7 to 11 cm central height; higher sides for side-sleeping

Study	Design and sample	Compared	Key result
Vanti 2019	Crossover RCT; 70 adults, chronic neck pain; 4 weeks; PEDro 8/10	Spring pillow vs education	Neck pain -8.7/100 (95% CI -14.7 to -2.6); headache -16.0; sleep quality no difference
Helewa 2007	Factorial RCT; 151 adults, chronic neck pain; 12-week primary endpoint	Pillow, exercise, both, placebo	Pillow alone no better than placebo; pillow + exercise best (NPQ 14.1 vs 18.6); interaction p = 0.029
Persson 1998	Randomized comparison; 55 participants; 6 neck pillows, 3 weeks	Shape and firmness	Preferred: soft, not too high, firm support under cervical curve; two-core design rated best
Sacco 2015	Within-subject EMG and comfort; 21 asymptomatic adults, side-lying	5 vs 10 vs 14 cm foam	10 cm most comfortable ($p < 0.001$); 5 cm highest trapezius activity
Lavin 1997	Crossover trial; 41 patients with cervical pain; PEDro 3/10	Water pillow vs roll vs usual	Water pillow: less morning pain, better sleep; roll pillow shortened sleep
Lei 2021	Narrative review of height determinants and test methods	Supine and lateral heights	Supine 7 to 10 cm, lateral about 10 cm in individual studies; no firm consensus

3. The pillow this evidence describes

Specification, no extras

- **Fill:** one solid block of natural latex (not shredded, not "latex blend" with polyester). It is the only material with repeated positive trial results. Dense viscoelastic (memory) foam is a reasonable second choice but was tested only as "foam" in the trials and showed no advantage over people's own pillows.
- **Height:** about 10 cm when your head is resting on it if you sleep on your side; 7 to 10 cm if you sleep on your back. Check it with the side-lying test below rather than the number on the box.
- **Firmness:** medium: it should support the neck without the head sinking to the mattress, and should not push the head up.
- **Shape:** plain rectangular or a gentle neck-support curve. A moulded contour added nothing in the trials.
- **Skip:** feather/down, very low (5 cm) pillows, roll pillows, and "cooling gel" or copper layers: the first two were worse in trials; the rest have no neck-pain evidence at all.
- **Trial it honestly:** trials used 1 to 4 weeks per pillow. Give a new pillow at least two weeks; most retailers of latex pillows allow this.

How to check the height at home (side sleepers)

- Lie on your side on your usual mattress with the pillow under your head only, shoulders on the mattress.
- Have someone look from the foot of the bed: your nose, chin and sternum should line up; the neck should be level with the mattress, not tilted up or down.
- If the head tilts down toward the mattress, the pillow is too low (the trial pattern that raised trapezius activity); if it tilts up, too high.
- Back sleepers: the chin should not be pushed toward the chest; the forehead and chin should be roughly level.

What people with neck pain can reasonably expect

- A better pillow mainly changes **waking** symptoms (morning stiffness, headache, arm/shoulder ache); it had no measurable effect on sleep quality in pooled data.

- Effect sizes are small to moderate: about 9 points on a 100-point pain scale in the spring-pillow RCT, and a halved risk of waking headache with latex versus polyester.
- The largest improvement in the literature came from combining a supportive pillow with a 6-week physiotherapist-led neck exercise program.
- If neck pain is new, worsening, follows trauma, or comes with arm weakness, numbness or night pain that is not positional, the pillow is not the problem; assess and refer.

4. Marketing claims against the literature

Claim	What the trials show
"Orthopedic contour / cervical pillow"	Contour foam = plain foam in the only head-to-head field trial. Support under the neck matters; the wave shape does not.
"Memory foam relieves neck pain"	Tested as "foam": performed like people's own pillows. Not harmful, not better than latex.
"Hotel-quality down"	Feather/down was the worst material in every comparison and the most abandoned.
"Cooling gel / graphite / copper"	No neck-pain trials. One review lists a cooling surface as a comfort factor only.
"Adjustable shredded fill"	Untested for neck pain; useful only in that it lets you reach the right height.
"Recommended by sleep doctors"	No sleep-medicine society guideline recommends a pillow or brand. The evidence is from physiotherapy and ergonomics trials, summarized above.

5. References

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Educational summary of published trials; not a substitute for individual assessment. Prices, warranties and return windows are deliberately omitted because they are not evidence.